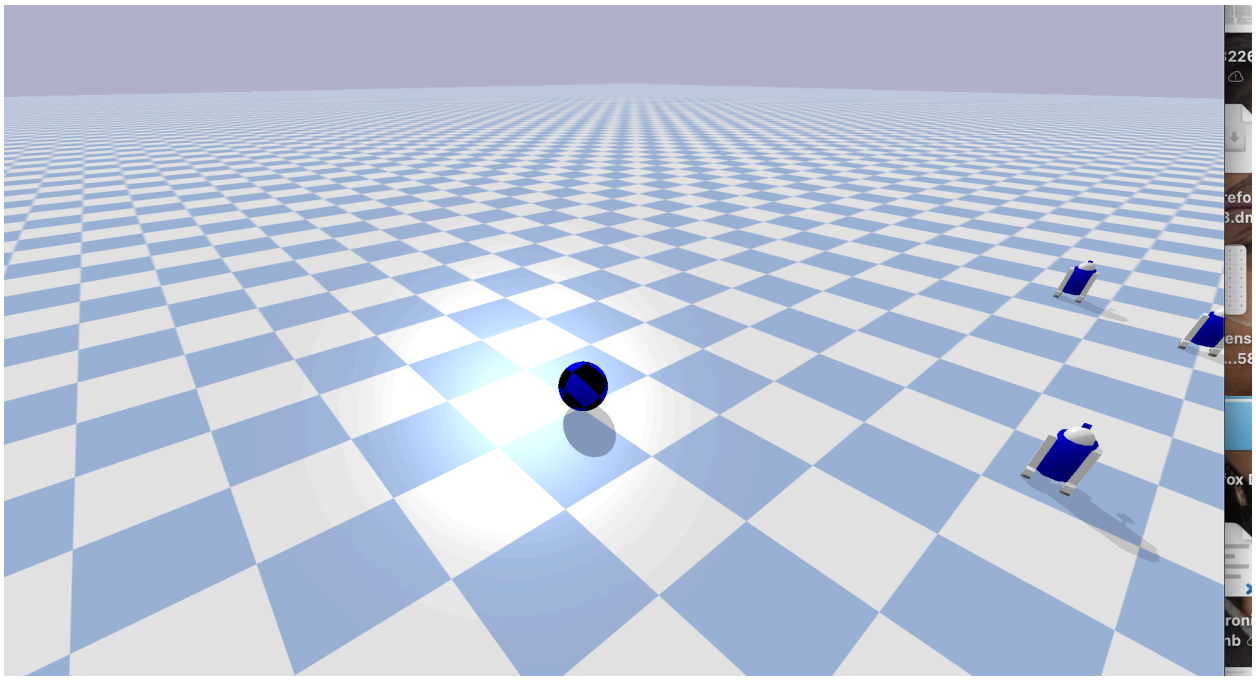
Visualization and Graphs



```
-----------------------------
| time/              |      |
|    fps             | 2893 |
|    iterations      | 1    |
|    time_elapsed    | 0    |
|    total_timesteps | 2048 |
-----------------------------
-----------------------------------------
| time/                   |             |
|    fps                  | 2269        |
|    iterations           | 2           |
|    time_elapsed         | 1           |
|    total_timesteps      | 4096        |
| train/                  |             |
|    approx_kl            | 0.014654114 |
|    clip_fraction        | 0.0908      |
|    clip_range           | 0.2         |
|    entropy_loss         | -1.38       |
|    explained_variance   | -0.000567   |
|    learning_rate        | 0.0003      |
|    loss                 | -0.0294     |
|    n_updates            | 10          |
```

```
|    policy_gradient_loss | -0.00251    |
|    value_loss           | 0.0885      |
-----------------------------------------
------------------------------------------
| time/                   |             |
|    fps                  | 2134        |
|    iterations           | 3           |
|    time_elapsed         | 2           |
|    total_timesteps      | 6144        |
| train/                  |             |
|    approx_kl            | 0.0069417516 |
|    clip_fraction        | 0.0345      |
|    clip_range           | 0.2         |
|    entropy_loss         | -1.36       |
|    explained_variance   | -2.03e-06   |
|    learning_rate        | 0.0003      |
|    loss                 | 0.0151      |
|    n_updates            | 20          |
|    policy_gradient_loss | -0.000629   |
|    value_loss           | 0.069       |
------------------------------------------
-----------------------------------------
| time/                   |            |
|    fps                  | 2065       |
|    iterations           | 4          |
|    time_elapsed         | 3          |
|    total_timesteps      | 8192       |
| train/                  |            |
|    approx_kl            | 0.005249697 |
|    clip_fraction        | 0.0715     |
|    clip_range           | 0.2        |
|    entropy_loss         | -1.33      |
|    explained_variance   | 1.55e-06   |
|    learning_rate        | 0.0003     |
|    loss                 | -0.00934   |
|    n_updates            | 30         |
|    policy_gradient_loss | -0.00145   |
|    value_loss           | 0.0436     |
-----------------------------------------
-----------------------------------------
| time/                   |            |
|    fps                  | 2032       |
|    iterations           | 5          |
|    time_elapsed         | 5          |
|    total_timesteps      | 10240      |
| train/                  |            |
|    approx_kl            | 0.018550687 |
|    clip_fraction        | 0.224      |
|    clip_range           | 0.2        |
```

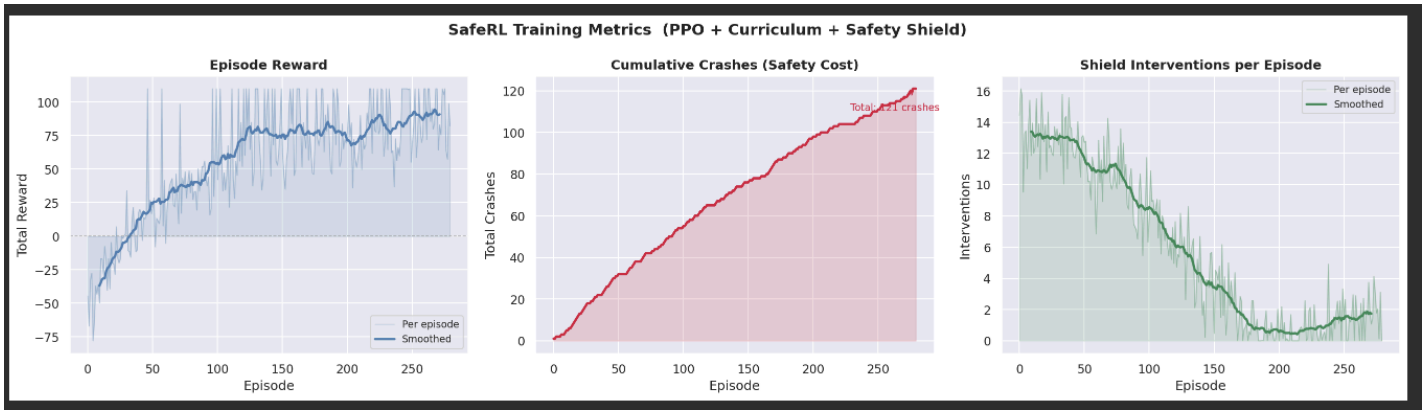
```
|    entropy_loss         | -1.3        |
|    explained_variance   | 1.67e-06    |
|    learning_rate        | 0.0003      |
|    loss                 | 0.0244      |
|    n_updates            | 40          |
|    policy_gradient_loss | -0.0065     |
|    value_loss           | 0.0286      |
-----------------------------------------
-----------------------------------------
| time/                   |             |
|    fps                  | 2007        |
|    iterations           | 6           |
|    time_elapsed         | 6           |
|    total_timesteps      | 12288       |
| train/                  |             |
|    approx_kl            | 0.0138030285 |
|    clip_fraction        | 0.133       |
|    clip_range           | 0.2         |
|    entropy_loss         | -1.23       |
|    explained_variance   | 2.74e-06    |
|    learning_rate        | 0.0003      |
|    loss                 | -0.0246     |
|    n_updates            | 50          |
|    policy_gradient_loss | -0.00238    |
|    value_loss           | 0.0203      |
-----------------------------------------
-----------------------------------------
| time/                   |             |
|    fps                  | 1991        |
|    iterations           | 7           |
|    time_elapsed         | 7           |
|    total_timesteps      | 14336       |
| train/                  |             |
|    approx_kl            | 0.008999315 |
|    clip_fraction        | 0.0675      |
|    clip_range           | 0.2         |
|    entropy_loss         | -1.21       |
|    explained_variance   | -5.96e-07   |
|    learning_rate        | 0.0003      |
|    loss                 | -0.0231     |
|    n_updates            | 60          |
|    policy_gradient_loss | -0.00347    |
|    value_loss           | 0.0133      |
-----------------------------------------
-----------------------------------------
| time/                   |             |
|    fps                  | 1965        |
|    iterations           | 8           |
|    time_elapsed         | 8           |
```

```
|    total_timesteps      | 16384        |
| train/                  |              |
|    approx_kl            | 0.0068799406 |
|    clip_fraction        | 0.038        |
|    clip_range           | 0.2          |
|    entropy_loss         | -1.14        |
|    explained_variance   | 1.79e-07     |
|    learning_rate        | 0.0003       |
|    loss                 | -0.0168      |
|    n_updates            | 70           |
|    policy_gradient_loss | -4.11e-05    |
|    value_loss           | 0.00962      |
-----------------------------------------
-----------------------------------------
| time/                   |             |
|    fps                  | 1931        |
|    iterations           | 9           |
|    time_elapsed         | 9           |
|    total_timesteps      | 18432       |
| train/                  |             |
|    approx_kl            | 0.009867953 |
|    clip_fraction        | 0.085       |
|    clip_range           | 0.2         |
|    entropy_loss         | -1.11       |
|    explained_variance   | 1.67e-06    |
|    learning_rate        | 0.0003      |
|    loss                 | -0.017      |
|    n_updates            | 80          |
|    policy_gradient_loss | -0.000918   |
|    value_loss           | 0.00638     |
-----------------------------------------
-----------------------------------------
| time/                   |             |
|    fps                  | 1923        |
|    iterations           | 10          |
|    time_elapsed         | 10          |
|    total_timesteps      | 20480       |
| train/                  |             |
|    approx_kl            | 0.014123214 |
|    clip_fraction        | 0.051       |
|    clip_range           | 0.2         |
|    entropy_loss         | -1.11       |
|    explained_variance   | 1.85e-06    |
|    learning_rate        | 0.0003      |
|    loss                 | 0.0131      |
|    n_updates            | 90          |
|    policy_gradient_loss | -0.00331    |
|    value_loss           | 0.00442     |
-----------------------------------------
```

```
-----------------------------------------
| time/                   |             |
|    fps                  | 1910        |
|    iterations           | 11          |
|    time_elapsed         | 11          |
|    total_timesteps      | 22528       |
| train/                  |             |
|    approx_kl            | 0.009070042 |
|    clip_fraction        | 0.0247      |
|    clip_range           | 0.2         |
|    entropy_loss         | -1.11       |
|    explained_variance   | -7.15e-07   |
|    learning_rate        | 0.0003      |
|    loss                 | -0.00713    |
|    n_updates            | 100         |
|    policy_gradient_loss | 0.000458    |
|    value_loss           | 0.00291     |
-----------------------------------------
----------------------------------------
| time/                   |            |
|    fps                  | 1904       |
|    iterations           | 12         |
|    time_elapsed         | 12         |
|    total_timesteps      | 24576      |
| train/                  |            |
|    approx_kl            | 0.01725798 |
|    clip_fraction        | 0.158      |
|    clip_range           | 0.2        |
|    entropy_loss         | -1.02      |
|    explained_variance   | -1.43e-06  |
|    learning_rate        | 0.0003     |
|    loss                 | 0.0191     |
|    n_updates            | 110        |
|    policy_gradient_loss | -0.00115   |
|    value_loss           | 0.00184    |
----------------------------------------
----------------------------------------
| time/                   |            |
|    fps                  | 1900       |
|    iterations           | 13         |
|    time_elapsed         | 14         |
|    total_timesteps      | 26624      |
| train/                  |            |
|    approx_kl            | 0.01396133 |
|    clip_fraction        | 0.153      |
|    clip_range           | 0.2        |
|    entropy_loss         | -0.923     |
|    explained_variance   | 7.75e-07   |
|    learning_rate        | 0.0003     |
```

```
|    loss                 | -0.00481   |
|    n_updates            | 120        |
|    policy_gradient_loss | -0.000617  |
|    value_loss           | 0.00116    |
-----------------------------------------
------------------------------------------
| time/                   |             |
|    fps                  | 1897        |
|    iterations           | 14          |
|    time_elapsed         | 15          |
|    total_timesteps      | 28672       |
| train/                  |             |
|    approx_kl            | 0.011980173 |
|    clip_fraction        | 0.128       |
|    clip_range           | 0.2         |
|    entropy_loss         | -0.831      |
|    explained_variance   | -1.22e-05   |
|    learning_rate        | 0.0003      |
|    loss                 | 0.0248      |
|    n_updates            | 130         |
|    policy_gradient_loss | -0.000586   |
|    value_loss           | 0.000925    |
------------------------------------------
-------------------------------------------
| time/                   |              |
|    fps                  | 1894         |
|    iterations           | 15           |
|    time_elapsed         | 16           |
|    total_timesteps      | 30720        |
| train/                  |              |
|    approx_kl            | 0.0061080013 |
|    clip_fraction        | 0.0294       |
|    clip_range           | 0.2          |
|    entropy_loss         | -0.728       |
|    explained_variance   | -5.13e-06    |
|    learning_rate        | 0.0003       |
|    loss                 | -0.00956     |

|    n_updates            | 140          |
|    policy_gradient_loss | 0.00168      |
|    value_loss           | 0.000626     |
-------------------------------------------
```


SafeRL Training Metrics (PPO + Curriculum + Safety Shield)
Episode Reward
Total Reward
Episode
Per episode
Smoothed
Cumulative Crashes (Safety Cost)
Total Crashes
Episode
Total: 211 crashes
Shield Interventions per Episode
Interventions
Episode
Per episode
Smoothed